



EDA-Based Analysis of Online Shopping Trends



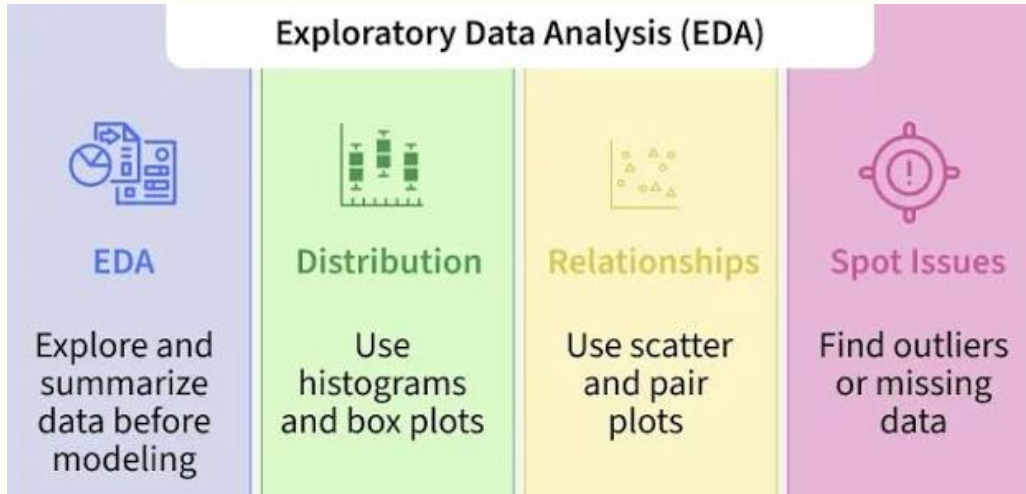
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Fundamentals of Data Science

INTRODUCTION

- Online shopping has become an essential part of modern consumer life, generating large amounts of data through customer purchases, product preferences, payment methods, and transaction patterns.
- Exploratory Data Analysis (EDA) helps transform this raw data into meaningful insights by identifying trends, relationships, and patterns in customer behavior.
- This project analyzes online shopping data using statistical methods and visualizations to understand purchasing trends, popular product categories, customer preferences, and sales patterns.
- The findings can support businesses in making better data-driven decisions and improving customer experiences.

KEY ACTIVITIES



WHAT ARE WE LOOKING FOR?

- When do people shop more frequently?**
 - Identify monthly, seasonal, or time-based purchasing patterns.
- How much do customers spend at maximum?**
 - Explore order values, spending distributions, and high-value purchases.
- How do people pay and in what way?**
 - Compare different payment preferences.
- What do customers prefer and like?**
 - Analyze ratings, reviews, and purchasing behavior.

WHAT DOES THE DATA TELL US?



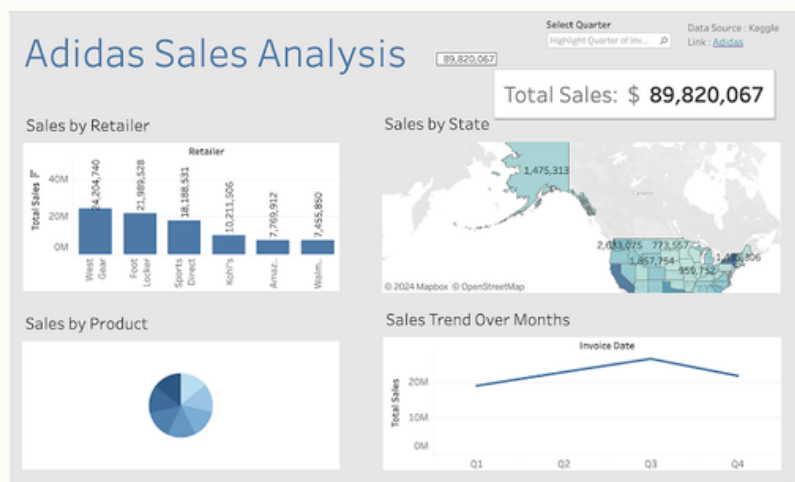
- Every click, search, order, rating, and payment creates valuable data.
- But raw data alone doesn't tell the story.
- Exploratory Data Analysis (EDA) helps uncover that story by revealing what customers buy, when they shop, how much they spend, and what influences their choices.

WHY DOES IT MATTER?

Understanding shopping behavior can help businesses:

- Target customers better
- Improve product selection
- Increase sales
- Improve customer experience
- Make smarter business decision

SAMPLE EDA OF ADIDAS



EDA PROCESS

- Collect** – Gather shopping data
- Clean** – Handle missing & duplicate data
- Explore** – Find patterns and trends
- Visualize** – Create meaningful graphs
- Discover** – Identify key insights
- Interpret** – Support data-driven decisions

BUSINESS IMPACT

- Predict Demand:** Use past shopping patterns to identify products with potentially higher demand.
- Optimize Strategy:** Use customer insights to improve pricing, promotions, and inventory planning.
- Improve Experience:** Understand customer preferences to offer more relevant products and personalized shopping experiences.

KEY INSIGHTS

- Customer Behavior:** EDA reveals purchasing patterns, spending habits, preferred product categories, and payment choices, helping understand what customers actually prefer.
- Sales Trends:** Visual analysis can uncover high-demand products, peak shopping periods, seasonal variations, and changes in purchase volume over time.
- Business Insights:** These patterns can help businesses improve product recommendations, plan inventory, target customers more effectively, and make data-driven decisions.